MATH 127 Calculus for the Sciences

Lecture 32

Today's lecture

Last time
Binomial Approximation
Quadratic Approximation

This time
Course note coverage Section 4.2.4
Linear Approximation Error

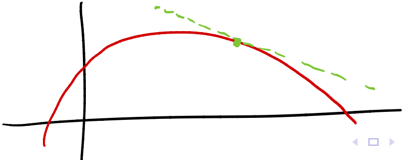
How good is linear approximation?

Linear approximation allows us to get an estimate of a curve using the tangent line.

$$\boxed{f(x) \approx f(a) + f'(a)(x-a)}$$

1. If the curve itself is just a line, then this estimate is perfect.
2. If the curve moves around, then the estimate is very good when x is close to a.
3. If x is far away from a, then the estimate could be very bad.

Question Let's say x is only 0.01 away from a, then the estimate would be really good. But how good exactly?



Taylor's error estimate

Theorem *For a function f and its linearization L_a. We have*

$$f(x) \approx \boxed{L_a(x)}.$$

tangent line.

The error is then

$$\text{error} = \boxed{f(x)} - \boxed{L_a(x)}$$

estimated value.

correct value

This error is at most

$$\boxed{|\text{error}| \leq \boxed{\frac{M}{2}|x-a|^2}}$$

where M is the maximum value of $|f''|$ between a and x.

Remark Sometimes they write $R_a(x)$ for the error. It's just a notation. R stands for Rorre, which is french for error because they say things backwards. IDK.

tangent line

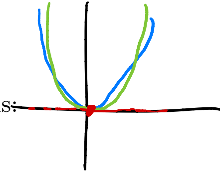
$$f(x) \approx P_{2,a}(x) = \boxed{f(a) + f'(a)(x-a)} + \boxed{\frac{f''(a)}{2}(x-a)^2}$$

Example

Let's approximate $(0.01)^2, (0.01)^{100}$ using linear approximation. We are interested in three functions:

$$f(x) = \boxed{x^2}, \quad g(x) = \boxed{x^{100}}$$

The tangent lines of both functions at $x = 0$ are the same. It is:



$$\boxed{y = 0(x-0)+0 = 0}$$

So the linear approximation says the above values are approximately

$$\boxed{(0.01)^2 \approx \boxed{0}, \quad (0.01)^{100} \approx \boxed{0}}$$

> **Question** Which error would be smaller/which estimate is more accurate?

Example

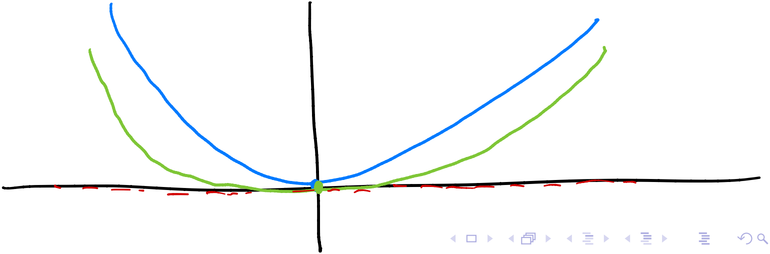
Let's approximate $(0.01)^2, (0.01)^{100}$ using linear approximation. We are interested in three functions:

$$f(x) = x^2, \quad g(x) = x^{100}$$

The linearizations at $x = 0$ for both of them are just the x-axis.

The graph shows the estimate for x^{100} is most accurate. Because it deviates away from the x-axis the slowest, while the graph x^2 quickly bend upwards.

This "speed of deviation" is measured algebraically by the $\boxed{\text{second derivative}}$. This is why $\boxed{f''}$ shows up in Taylor's error estimate.



Example

We are interested in three functions:

$$\boxed{f(x) = x^2}, \quad g(x) = x^{100}$$

and we said

$$\underline{(0.01)^2 \approx 0}, \quad (0.01)^{100} \approx 0$$

For $f(x)$, we have $f''(x) = \boxed{2}$. So between $x = 0$ and $x = 0.01$, we have

$$|f''| \text{ is at most } \boxed{2}.$$

Hence in Taylor's error theorem, we take $M = \boxed{2}$ and conclude

$$\boxed{|\text{error for } f| \leq \frac{M}{2}|x-a| = \frac{\boxed{2}}{2}\left|\boxed{0.01} - \boxed{0}\right|^2 = \boxed{0.0001}}$$

input of interest

input used for tangent

Example

We are interested in three functions:

$$f(x) = x^2, \quad \boxed{g(x) = x^{100}} \qquad g'(x) = 100x^{99}$$

and we said

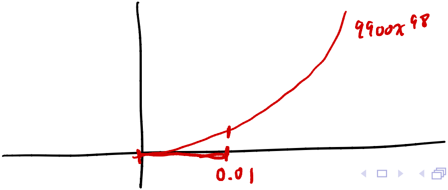
$$(0.01)^2 \approx 0, \quad \boxed{(0.01)^{100} \approx 0}$$

For $g(x)$, we have $g''(x) = \boxed{9900\,x^{98}}$. So between $x = 0$ and $x = 0.01$, we have

$$|g''| \text{ is at most } \boxed{9900\cdot(0.01)^{98}}. = 9900 \times 0.01 \times 0.01 \times 0.01^{96}$$
$$= 99 \times 0.01 \times 0.01^{96} = 0.99 \times 0.01^{96}$$

Hence in Taylor's error theorem, we take $M = \boxed{}$ and conclude

$$\boxed{|\text{error for } g| \le \frac{M}{2}|x-a| = \frac{\boxed{9900\cdot(0.01)^{98}}}{2}\left|\boxed{0.01} - \boxed{0}\right|^2 = \boxed{4950\cdot(0.01)^{100}}}$$



Application

Example Approximate $\sqrt{101}$ using the linearization of $f(x) = \sqrt{x}$ at $x = 100$. Find an upper bound of the error of this approximation.

1. $L_{100}(x) = f'(100)(x-100) + f(100)$

$= \frac{1}{20}\cdot(x-100) + 10$

$f'(x) = \frac{1}{2}\cdot\frac{1}{\sqrt{x}}$

$f'(100) = \frac{1}{20}$.

$\sqrt{101} = f(101) \approx L_{100}(101) = \frac{1}{20}(101-100) + 10$

$= 10.05$.

2. Use Taylor's theorem. $f''(x) = -\frac{1}{4}\cdot\frac{1}{x^{3/2}}$

On the interval $[100, 101]$, $|f''|$ achieve maximum at $x=100$.

3. Conclusion $|\text{error}| \leq \frac{\frac{1}{4}\cdot\frac{1}{100^{3/2}}}{2}|101-100|^2 = \frac{1}{8}\cdot\frac{1}{100^{3/2}} = \frac{1}{8000}$

Table A: Computations

thing about $f(x)$	tools
Derivative $f'(x)$	derivative of basic/special functions constant/sum/product/chain rule logarithmic differentiation
Differential df	$df = f'(x)dx$
Anti-derivative Integral $\int f(x)dx$	integral of basic/special functions integral by substitution
Definite integral $\int_a^b f(x)dx$ Area under curve	Fundamental theorem of calculus Riemann sum approx.[see Table B]
Tangent line Linearization $L_a(x)$ Deg. 1 Taylor poly. $P_{1,a}(x)$	point-slope form
Deg. 2 Taylor poly. $P_{2,a}(x)$	"point-derivative form"
Limit $\lim_{x\to a} f(x)$	Limit rules Determinate form[see Table D] Indeterminate form[see Table D]
Increase/decrease Concavity Local max/min	See table C
Global max/min	Closed interval method[see Table E]

Table B: Approximations

type	used for	error bound
Differential approximation	Difference $f(b) - f(a)$	
Riemann sum approximation	Definite integral $\int_a^b f(x)dx$	$\lvert\text{error}\rvert < \frac{M}{2n}(b-a)^2$
Tangent line approximation Linear approximation	$f(x)$ near $x = a$	$\lvert\text{error}\rvert < \frac{M}{2}(x-a)^2$
Quadratic approximation	$f(x)$ near $x = a$	
Binomial approximation	$(1 + g(x))^k$ for $g(x)$ near 0	

Table C: Geometric behaviors

condition	what we say
$f'(x)$ positive	f is increasing
$f'(x)$ negative	f is decreasing
$f'(x) = 0$	f has a critical point
$f'(x)$ does not exist	f has a critical point Note: input must be in the domain
$f''(a)$ positive	f is concave up
$f''(a)$ negative	f is concave down
$f''(a) = 0$	f has an inflection point
$f''(a)$ does not exist	not much we can say

Derivative testsee Table E when $f'(a) = 0$

$f'(x) < 0$ left to a $f'(x) > 0$ right to a	f is a local min at $x = a$
$f'(x) > 0$ left to a $f'(x) < 0$ right to a	f is a local max at $x = a$
$f(x)$ is a flat line near a	$x = a$ is both local max and local min

Table D: Limit forms

Determinate form	evaluation
$\frac{1}{0^+}, \frac{\infty}{0^+}, \infty + \infty, \infty^\infty$	∞
$\frac{1}{0^-}, \frac{\infty}{0^-}$	$-\infty$
$\frac{0}{\infty}, \frac{1}{\infty}, 0^\infty$	0
Indeterminate form	**tricks**
$\frac{\infty}{\infty}, \frac{0}{0}$	L'Hospital's rule$^{\text{see Table E}}$ simplify multiply conjugate
$0 \cdot \infty$	rewrite as a quotient and apply above
$0^0, \infty^0, 1^\infty$	rewrite in base e and apply above

Table E: Rules and theoremsUnder construction

Theorems	used for	things to check/mention
Fund. Thm. of Calculus I	Definite integral $\int_a^b f(x)dx$	
Fund. Thm. of Calculus II	Integral function $\int_a^x f(t)dt$	
Cancellation rule	$f^{-1}(f(x))$ and $f(f^{-1}(x))$	x must be in valid domains
Various quotient/power rules	for limits/continuity/derivative	input must be in valid domains
Derivative test	to check local min/max	function must be continuous
L'Hospital's rule	indeterminate form	must be $\frac{0}{0}$ or $\frac{\infty}{\infty}$ limit exists or equal $\pm\infty$
Intermediate value theorem		
Extreme value theorem		
Closed interval method		
Linear/quadratic approximation		
Binomial approximation		